

CarPlay must have up to 48kHz, 16-bit stereo output, and 24kHz, 16-bit mono input from the microphone. Alternate audio output is always mixed with the main audio output and is used for notification prompts. This channel can be 44.1kHz or 48kHz, stereo or mono, and deliver 16-bit audio.

For user input to interact with the CarPlay interface, a car must have a touchscreen display, have physical controls that add redundancy to the system and give a secondary way of communication, and must also have a microphone that can be activated by a button to receive voice input. The touchscreen display has to allow swipes for scrolling, with less than 140ms of touch latency. There is alternate compatibility for low resolution, resistive touchscreen displays as well, although Apple does not recommend it. This CarPlay interface is optimised for single touch operations.

As for physical input, the buttons in the car's system must include a rotating knob for scrolling, a selection button and a back button. Optional additions may also include a multi-way rotating selector (like the one used by Mercedes-Benz), gesture recognition for character input (like the BMW iDrive controller), and dedicated music control and telephony keys. Other recommendations by Apple include a speed sensor to relay motion data, and integrated GPS module for wireless Apple CarPlay connections to gather location data from.

For all systems to connect physically, a car needs to have a USB Type-A port. For wireless connections, the car must have both Bluetooth and WLAN support. The Wi-Fi connection must be established via a 5GHz, 802.11ac Wi-Fi modem. For the entire list of instructions set forth by Apple, head to <http://digit.in/carplycomp>



Any iPhone with a Lightning port right now supports Apple CarPlay